MILITARY SPECIFICATION SHEET

ELECTRON TUBE, MAGNETRON

TYPE 7006

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Pulsed, tunable frequency range 9,000 to 9,600 MHz, rated peak power output 210 kw, with integral magnet

ABSOLUTE RATINGS:

Parameter:	Ef	If	if(surge)	tk	Du	tpc	rrv	epy	ib	Pi	pi
Unit:	V	A	a	sec	---	μs	kv μs	kv	a	W	kw
Maximum:	15.3	3.75	12	---	.0013	3.30	230	23	30	690	690
Minimum:	---	---	---	120	.0001	0.10	---	---	15	---	---

ABSOLUTE RATINGS:

Parameter:	VSWR	T(anode)	T(tuner)	(cathode T terminal)	Torque (Tuner drive)	Pressurization Input	Pressurization Output	Alt
Unit:	---	C	°C	°C	in.-oz	psia	psia	ft
Maximum:	1.5:1	+150	+150	+165	200	45	45	10,000
Minimum:	---	-55	-55	-55	---	40	40	---

PHYSICAL CHARACTERISTICS:

Dimensions:	See figure 1	Tuner dial marking:	See note 6
Mounting position:	Any	Cathode:	Unipotential
Mounting support:	Base plate	Magnet isolation:	See note 4
Weight:	11 pounds (approx)	Handling:	See notes 4 and 5
Output coupling:	To UG-52B U choke flange (modified)		

TEST CONDITIONS:

Parameter:	Ef	Du	tpc	rrv	Ib	VSWR
Unit:	V	---	μs	kv μs	mAdc	---
Tolerance:	---	---	---	min	---	max
Test condition 1:	9.5	.00018	0.12 ±0.02	220	5.0	1.1:1
Test condition 2:	0	.001	1.00 ±0.1	220	27.5	1.1:1

Frequency	
F	MHz
F1	9,000
F2	9,300
F3	9,600

GENERAL:

First article testing - See note 24

Preheating - Test condition 1

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METHOD	REQUIREMENT OR TEST	NOTES	TEST	CONDITIONS	SYMBOL	LIMITS		UNIT
						MINIMUM	MAXIMUM	
	Quality conformance inspection, part 1	1	-					
4315	Starting stability	7	1	tb 30 a; F F2	MP	---	0.5	
4003	Pressurizing	-	-	Input and output assemblies; Method A; duration 2 minutes; pressure 42.5 ± 2.5 psia	---	---	---	---
4223	Operating torque or force	-	-	No voltages; T (anode and tuner) 25° C	Torque	---	100	in.-oz
1301	Heater current	-	-	Ef = 13.75 V; tk = 300 sec (min)	If	3.00	3.75	A
4250	Output power measurement	-	1 2	t = 300 sec (min) F F1 to F3	Po Po	36 190	45 250	W W
4308	RF bandwidth	14 14, 15	1 2	F F1, F2, F3 F F1, F2, F3	BW BW	--- ---	2.0 tpc 2.5 tpc	MHz MHz
4308	Minor lobe ratio	-	1 2		Ratio Ratio	7 8	--- ---	dB dB
4315	Stability	16	1, 2	tb 30 a; VSWR 1.5:1	MP	---	0.25	
4310	Frequency pulling figure	-	1	F = F1, F2, F3	ΔF	---	15	MHz
4306	Pulse voltage	-	2	F F1, F2, F3	epv	19	23	kv
	Quality conformance inspection, part 2							
1031	Variable-frequency vibration	18, 23	-	No voltages	---	---	---	---
---	Variable-frequency vibration test end points:							
4308	RF bandwidth	14	1	F = F1, F2, F3	BW	---	2.0 tpc	MHz
4308	Minor lobe ratio	-	1		Ratio	7	---	dB
4315	Stability	16	1	tb = 30 a; VSWR = 1.5:1	MP	---	0.25	
---	Frequency drift	19	2		ΔF	---	25	MHz
4223	Operating torque or force	p, q (Fig 1)	-	T(anode) and T(tuner) = -55° and + 150° C	Torque	---	200	in.-oz
---	Backlash	20	2	F = F2; VSWR = 1.05:1	ΔF	---	10	MHz
4311	Frequency pushing figure	-	1	F = F1, F2, F3; Ib = 4.5 to 5.4 mAdc	ΔF	---	0.5	MHz a

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METHOD	REQUIREMENT OR TEST	NOTES	TEST	CONDITIONS	SYMBOL	LIMITS		UNIT
						MINIMUM	MAXIMUM	
	Quality conformance inspection, part 3							
1501	Intermittent life	3, 11, 12, 13, 21, 22	1, 2		Cycles	1,000	---	---
---	Intermittent life-test end points:							
4306	Pulse voltage	-	2		epv	19	23	kv
4250	Output power measurement	-	2		Po	150	---	W
4308	RF bandwidth	-	2		BW	---	3.0 tpc	MHz
4308	Minor lobe ratio	-	2		Ratio	-6	---	dB
4223	Mechanical tuning range	17, (r, q, Fig 1)	2		F	9,000	9,600	MHz
4315	Stability	-	2	ib 30 a	MP	---	0.5	%
	Periodic-check tests	2	-					
1002	Barometric pressure, reduced	-	1	Pressure 522 mmHg (absolute); VSWR 1.5:1 (vary phase ∧ 2)	---	---	---	---
1143	Forced cooling	p, q, r (Fig 1)	2	TA 25 ±3 C; Pi to Po 380 W (adjust ib); airflow 40 cfm (max)	T(rise)	---	50	°C
4309	Phase of sink	8	-	No voltages	---	0.40	0.45	deg
1041	Shock	-	-	No voltages; 15 G	---	---	---	---
---	Shock-test end points:							
4308	RF bandwidth	14	1	F F1, F2, F3	BW	---	2.0 tpc	MHz
4308	Minor lobe ratio	-	1		Ratio	7	---	dB
4315	Stability	16	1	ib 30 a; VSWR 1.5:1	MP	---	0.25	%
4027	Temperature coefficient	p, q (Fig 1)	2	F F2, T (anode) and T (tuner) 55 to 150 C	ΔF ΔT	---	0.25	MHz °C
4223	Mechanical tuning fatigue	9, 10	-	No voltages; T 25 ±3 C	---	5,000	---	Cycles

NOTES:

1. Unless otherwise specified, the AQL for all tests listed under quality conformance inspection, part 1, shall be 1.0, inspection level II.
2. If these tests have previously been performed by the tube manufacturer, and if the materials, manufacturing processes and test requirements are the same as when the tests were performed, the subject test (or tests) need not be performed.
3. The tube heater shall be protected against arcing by the use of a connector that places a minimum capacitance of 4,000 pF across the heater directly at the input terminals.
4. In handling and mounting the tube, care shall be exercised to prevent demagnetization. Ferromagnetic material shall not be brought within 12 inches of the tube; energized magnets shall not be brought within 18 inches of the tube.

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NOTES: -Continued

5. Reasonable care shall be used in transportation, storage, installation, and use of the tube to avoid the impartation of vibration or shock in excess of the values which the tube is designed to withstand.

6. The tuner dial settings referable to corresponding frequency values shall be marked on each tube, in successive 100 MHz steps of frequency. Calibration of the tuner dial shall be effected under conditions of T (anode) = 100° C (approx).

7. The starting stability test shall be performed on 100 percent of the submitted lot.

8. Measurement shall be made at the "hot test" frequency setting corresponding to 9,600 MHz (operational frequency), and shall be made using "cold test" techniques. The λg limits specified apply to the fraction of wavelength determinable as the test goal.

9. One cycle shall consist of a complete traverse from one end of the tuning range to the other end, and back again.

10. Separate samples shall be used for this test.

11. A modulator of the discharging-network type shall be used for test(s).

12. The standing-wave introducer shall be moved continuously, during the test, so that operation is obtained for all phases of VSWR. The standing-wave introducer shall be cycled continuously through a 1/2-wavelength of transmission line at a maximum rate of 4-cycles per hour.

13. The characteristics of the applied pulse shall be those which result in proper starting and oscillation. The rate of pulse voltage rise, the percentage of pulse voltage ripple, and the rate of pulse voltage fall are important considerations. Fall time of the current pulse shall not exceed 0.12 μs. No spike or ripple shall exceed ±7.0 percent of average peak of pulse current. Inverse voltage shall not exceed 30 percent of the forward voltage. There shall be no negative post-pulse voltage oscillations since they may result in leakage current and noise.

14. The tube shall be operated into a transmission line having a minimum VSWR of 1.5:1 adjusted in phase to produce maximum spectrum degradation.

15. The rf bandwidth shall be within the limit specified when a VSWR of 1.5:1 is introduced in the load at approximately 0.6 meters distance from the tube coupling flange, with the phase adjusted for widest spectrum.

16. With the VSWR adjusted to that phase producing maximum instability, the missing pulses shall be counted during the last 3 minutes of a test period not exceeding 5 minutes.

17. The tuner drive mechanism shall not be set against either mechanized stop.

18. The tube shall be fastened rigidly to the vibration platform and vibrated with a simple harmonic motion at F = 10 to 55 to 10 Hz in 1 to 3 minutes at 10 G or 0.060-inch double amplitude. This cycle shall be repeated for a total of 90 minutes in each of three mutually perpendicular planes, and then the tube shall be vibrated for 30 minutes at each of the resonant frequencies noted.

19. The frequency drift from start to stabilized frequency shall not be more than the specified amount, and at least one-half of the total frequency drift shall occur during the first 2 minutes of a 5-minute test period.

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NOTES: -Continued

20. After thermal equilibrium has been reached, the frequency obtained by turning the tuner from the beginning of the frequency range to the specified setting in one direction shall be reproducible within the specified limits when returning to that same setting from the extreme opposite direction.

21. Life test shall be run with heater power frequency set at the major resonant frequency, if any, as determined by the heater-structure test, provided that the resonant frequency occurs between 55 to 65 Hz or in the range between 340 to 1,800 Hz. If there is no such resonant frequency, heater power may be at any convenient frequency up to 1,800 Hz, at the discretion of the manufacturer.

22. A cycle for intermittent life-test operation shall consist of the following:

Test condition	Ib mAdc	Ef V	Minutes duration
Standby	0	13.75	2
1	5.0	9.5	5
2	27.5	0	40
Off	0	0	13

Operating frequencies shall be reset approximately every 24 hours as follows:

F = 9,000, 9,200, 9,400, 9,600 MHz; F = 9,000 MHz, etc.

23. This is a destructive test.

24. Unless otherwise stated in the contract or purchase order, requirements for first article inspection hereby replace those covering qualification approval specified herein. The testing program shall include all tests and requirements of this specification sheet and the applicable ones listed in table 1, samples and acceptance criteria for qualification of MIL-E-1. There are no allowable defects during first article testing. Requirements covering the first article test plan and test reports shall also be as specified in the contract or purchase order.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Army -
Air Force - 11, 17
DSA - ES

User activities:
Army - WC
Navy - AS, OS, MC, CG, SH

OTHER:
NSA-NS

Preparing activity: Army - EL

Agent: DSA - ES

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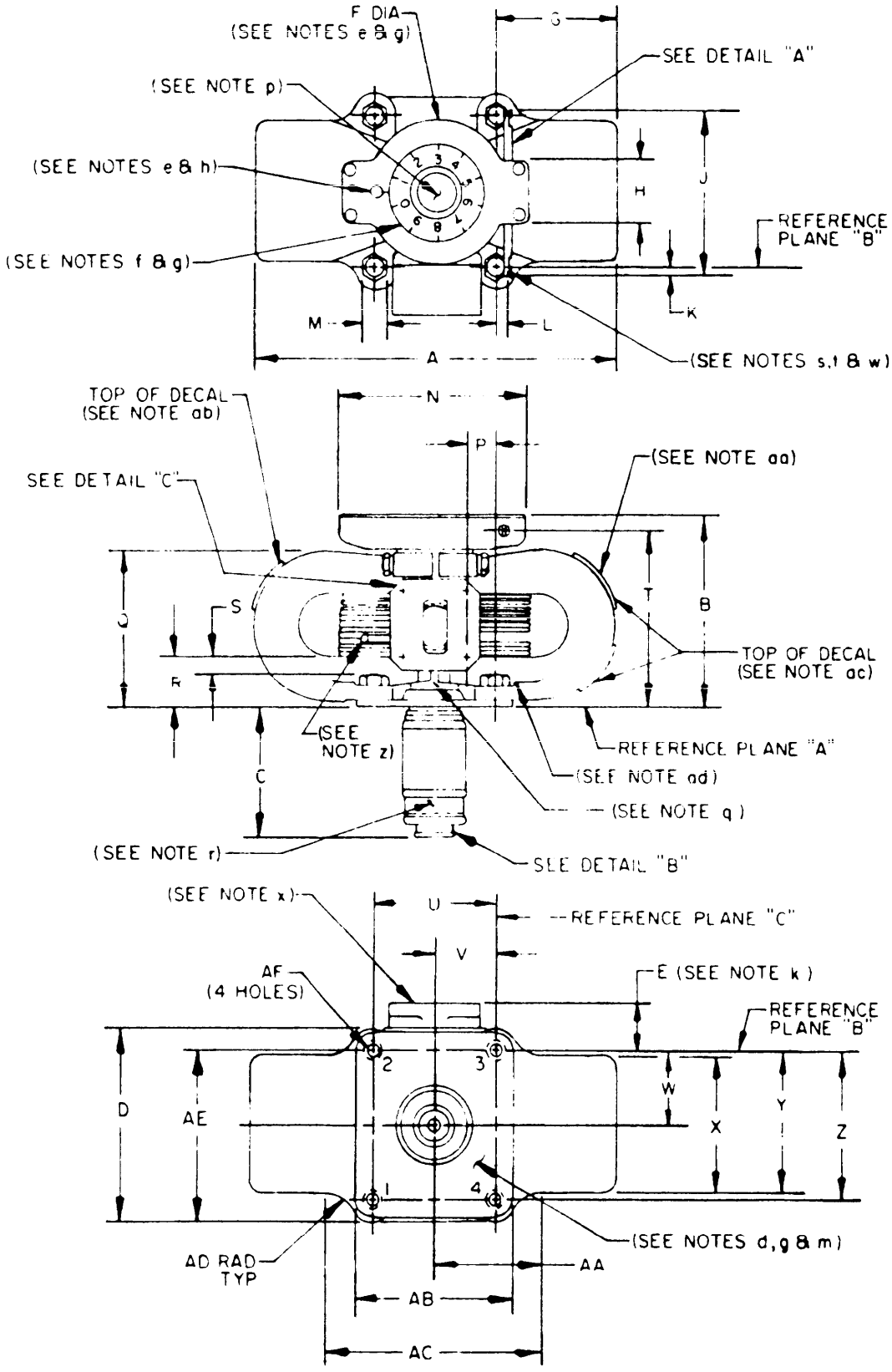


FIGURE 1. Outline drawing of electron tube type 7006.

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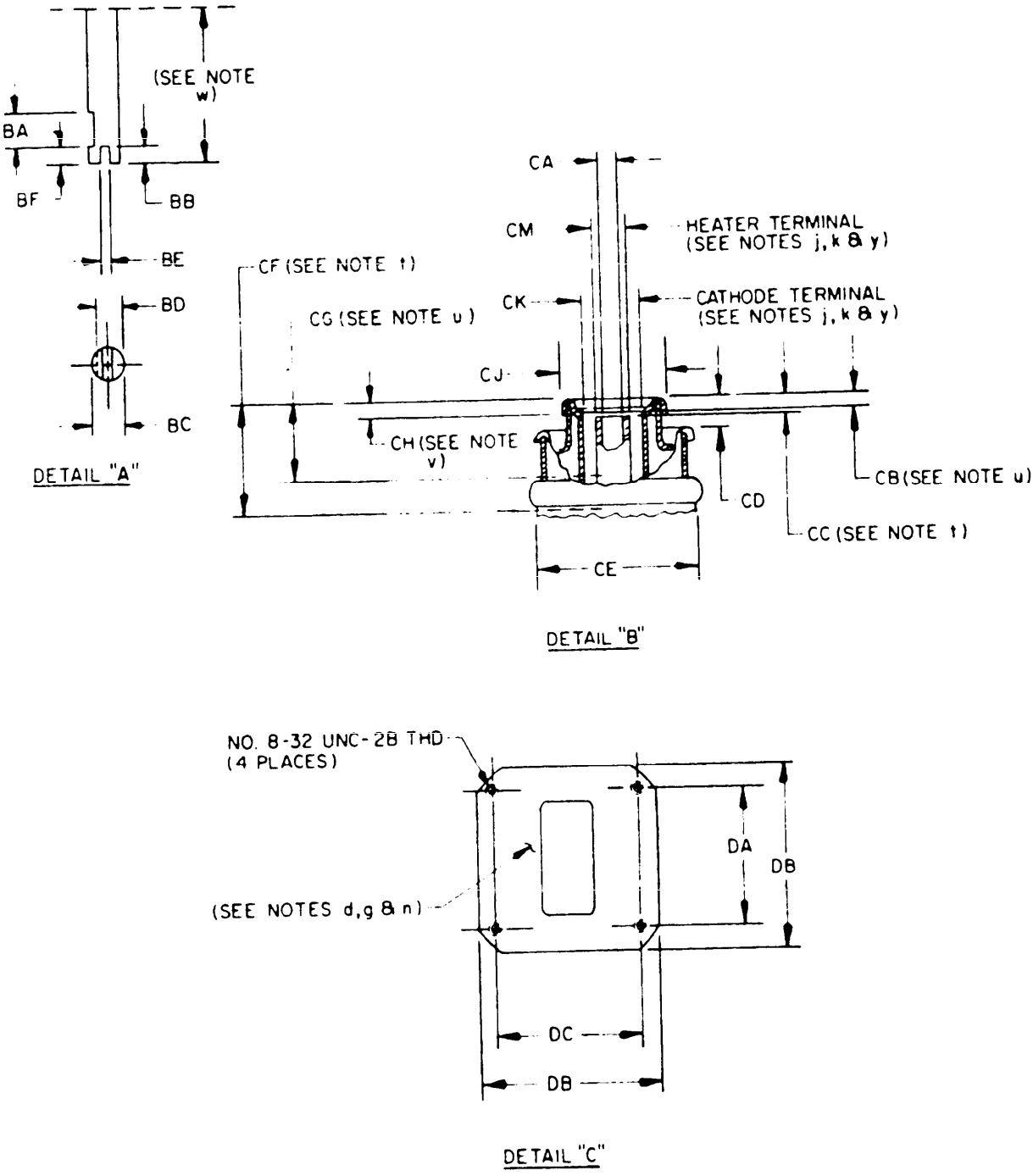


FIGURE 1. Outline drawing of electron tube type 7006 - Continued.

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 1		
A		7.718(196.04)
E	.882(22.40)	.932 (23.67)
K	.062 (1.57)	.187 (4.75)
L	.186 (4.72)	.256 (6.50)
P	.468(11.89)	.558 (14.17)
R	.945(24.00)	1.009 (25.63)
T	3.438(87.33)	3.530 (89.66)
AF	.276 (7.01)	.286 (7.26)
BE	.047 (1.19)	.057 (1.45)
CC	.125 (3.18)	.187 (4.75)
CJ	.825(20.96)	.838 (21.29)
CK	.540(13.72)	.545 (13.84)
DA	1.348(34.24)	1.356 (34.44)
DC	1.470(37.34)	1.478 (37.54)
Quality conformance inspection, part 2		
C	2.625(66.68)	2.750 (69.85)
G		2.607 (66.22)
J		3.250 (82.55)
Q		3.270 (83.06)
S	.593(15.06)	.656 (16.66)
U		2.500 (63.50)
Z		3.000 (76.20)
AA		2.350 (59.69)
AC		4.700(119.38)
AD		.960 (24.38)
CA	.164 (4.17)	.174 (4.42)
CD	.219 (5.56)	.281 (7.14)

Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 3		
B		3.937(100.00)
D		3.875 (98.43)
F		2.937Dia(74.60)
H		2.000 (50.80)
N		4.125(104.78)
X		2.875 (73.03)
Y		3.000 (76.20)
AB		3.468 (88.09)
AE		3.421 (86.89)
CB		.156 (3.96)
CE		1.500 (38.10)
CF	.750(19.05)	
CG	.316 (8.03)	
Reference dimensions		
M	.500(12.70)	
V	1.250(31.75)	
W	1.500(38.10)	
BA	.1875(4.76)	
BB	.125 (3.18)	
BC	.187 (4.75)	
BD	.1562Dia(3.97)	
BF	.125 (3.18)	
CH	.125 (3.18)	
CM	.250 (6.35)	
DB	1.830(46.48)	

FIGURE 1. Outline drawing of electron tube type 7006 - Continued.

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NOTES:

a. Ref. Plane "A" is a plane coincident with the indicated surface of the mounting flange.
b. Ref. Plane "B" is a plane perpendicular to Plane "A" and is coincident with the centers of mounting-flange holes No. 2 and No. 3.
c. Ref. Plane "C" is a plane perpendicular to Planes "A" and "B", and is coincident with the centers of mounting-flange holes No. 3 and No. 4.
d. These surfaces suitablefor hermetic-seal connections. Output coupling to mate with UG-52B/U choke flange modified by clearing threads.
e. The tuner housing shall have a scribed, black line from index hole to tuner dial.
f. The tuner dial face shall be black finish, and shall have 100 graduations scribed as follows: numbers 0 to 9 as indicated, and 1 10-division marks between the respective numbers. All numerals and division marks shall be white or silver color.
g. All metal surfaces shall be black finish, except mounting and connecting surfaces and tuner operators and indicators. Unpainted surfaces shall be nickel or silver plated, brass, or cadmium plated.
h. Number visible here shall be within the sequence of 0 to 6, and shall be indicative of the number of complete revolutions of the worm gear.
j. The heater and cathode terminals shall be concentric within .010(.25 mm). The axis of the heater-cathode terminal shall be within .046(1.17 mm) of specified location.
k. The limits include angular as well as lateral deviation.
m. All points on mounting surface (base plate) shall be within .015(.38 mm) of Plane "A".
n. With the flange on a plane surface, a .005(.13 mm) thickness gage, .125(3.18 mm) wide, shall not enter.
p. Tuner temperature measurement point.
q. Anode temperature measurement point.
r. Heater-cathode temperature measurement point.
s. Clockwise rotation of this end of tuner operator shall result in decreased frequency. Approximately 136 revolutions of tuner operator shall suffice to traverse the frequency range.
t. These dimensions define the limits of dimension CA.
u. These dimensions define the limits of dimension CK. The step with the dimension CB is optional.
v. No connector clamping means shall bear beyond this depth.
w. A sleeve .195(4.95 mm) I.D. by .406(10.31 mm) O.D. shall be able to pass over end of tuner operator to within .250(6.35 mm) of tuner housing.
x. A suitable dust cover to protect the opening of the waveguide output flange shall be provided with the tube.
y. Entire extent of connection areas of heater and cathode terminals shall be silver or nickel plated.
z. Seal-off shall not extend beyond outline of magnets.
aa. Frequency-dial setting information shall be located here.
ab. The following warning shall be located here: "WARNING: MAINTAIN MINIMUM CLEARANCE OF 2 INCHES BETWEEN MAGNET AND MAGNETIC MATERIAL".
ac. Identification markings (Serial No., date, etc.) shall be located here.
ad. Clearance above mounting-surface bulges shall be provided to enable installation of four 1.750-in.(44.45 mm) mounting bolts.

FIGURE 1. Outline drawing of electron tube type 7006 - Continued.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-E-1/1061D ELECTRON TUBE, MAGNETRON TYPE 7006

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

[ ] DIRECT GOVERNMENT CONTRACT [ ] SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

[ ] YES [ ] NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity).*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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